PT **Pushkaralu Crowd Digital Twin**

User journeys for all use cases

Step-by-step workflows with screenshots of the running application · generated 2026-10-23

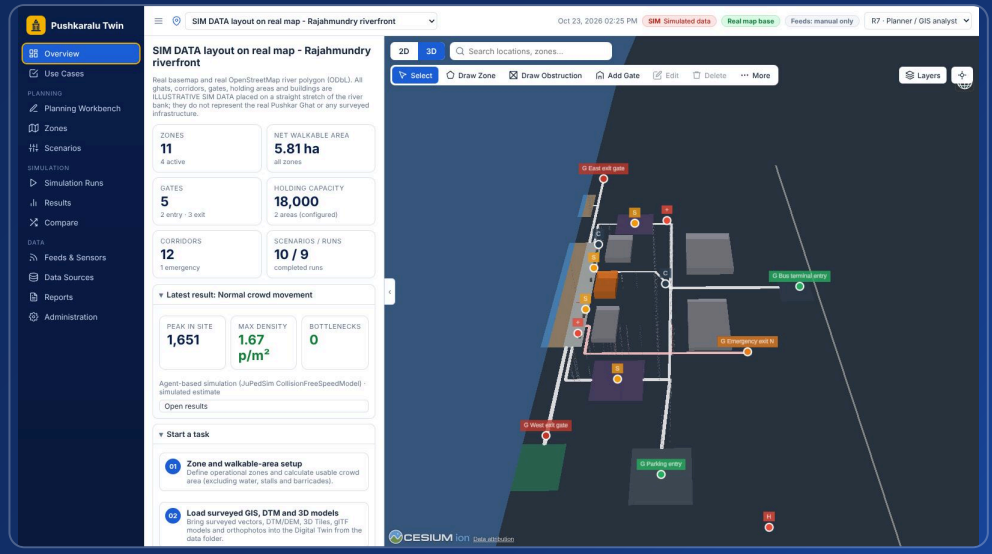
Use cases

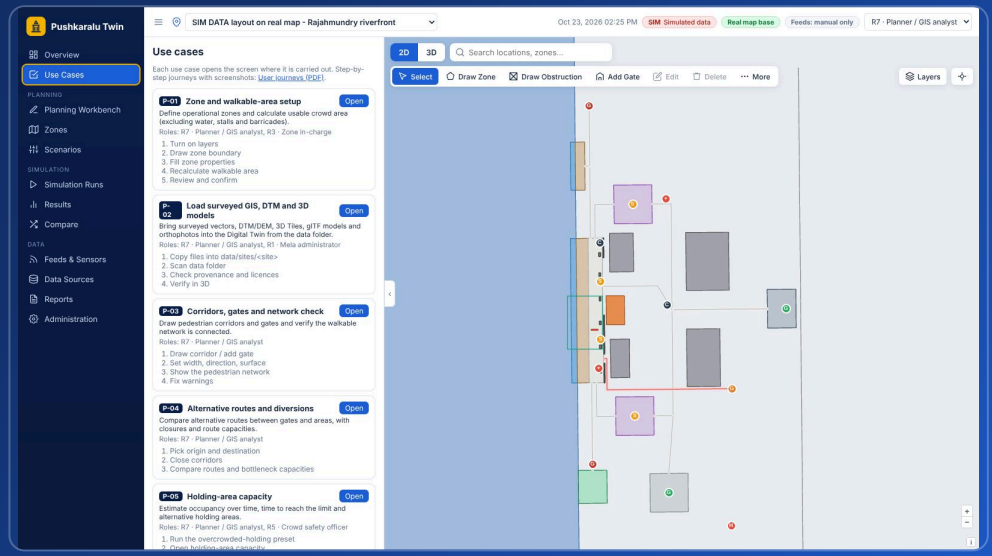
P-01	Zone and walkable-area setup	R7, R3
P-02	Load surveyed GIS, DTM and 3D models	R7, R1
P-03	Corridors, gates and network check	R7
P-04	Alternative routes and diversions	R7
P-05	Holding-area capacity	R7, R5
P-06	Scenario setup	R7
P-07	Run and monitor a simulation	R7
P-08	Density, queues and bottlenecks	R5
P-09	Blocked corridor and route failure	R7, R5
P-10	Emergency evacuation	R5, R1
P-11	Compare baseline and alternative	R5
P-12	Reports, SOP and drill package	R5, R1
P-13	Observed and forecast counts	R7
P-14	Density thresholds approval	R1, R5

Roles

- **R1** **Mela administrator** – Approves thresholds, SOPs and drill packages; activates zones.
- **R3** **Zone in-charge** – Reviews zone boundaries and owner departments; confirms and activates zones.
- **R5** **Crowd safety officer** – Runs and reviews density, bottleneck and evacuation analyses; reviews reports.
- **R7** **Planner / GIS analyst** – Prepares GIS data, zones, corridors and scenarios; runs simulations.

Roles are selected in the top bar; the prototype has no login, so approvals are records, not authenticated sign-off.





All screenshots use the bundled SIM DATA layouts (synthetic geometry; the Rajahmundry layout is illustrative and placed on the real OpenStreetMap river bank). Simulation results are estimates. Basemap tiles were not reachable in the environment where these screenshots were captured, so the map background appears plain; on a connected computer the selected real basemap is shown.

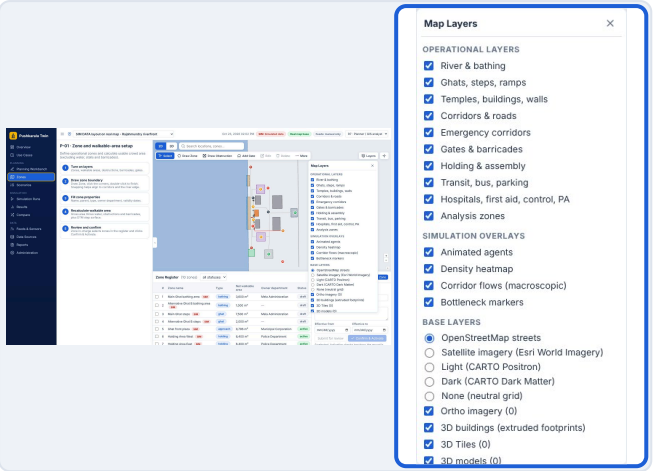
PT **Pushkaralu Twin**
User journey

P-01 Zone and walkable-area setup

Define operational zones and calculate usable crowd area (excluding water, stalls and barricades).

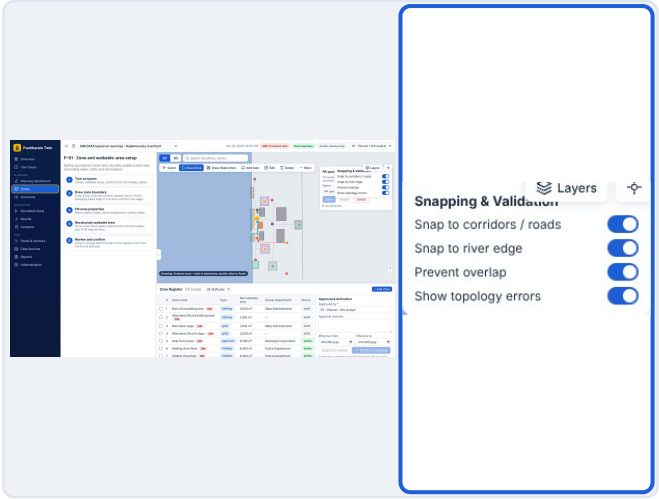
R7 · Planner / GIS analyst | R3 · Zone in-charge

1 Turn on layers



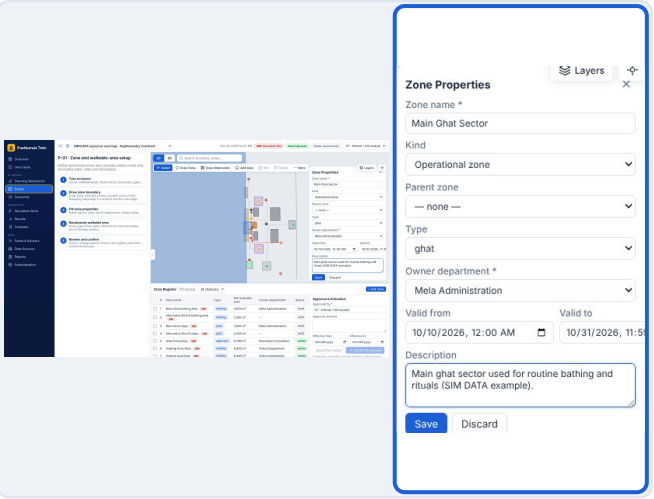
R7 opens Layers: operational layers (zones, walkable areas, obstructions, barricades, gates) and base layers (real map, ortho imagery, 3D buildings, DTM).

2 Draw zone boundary



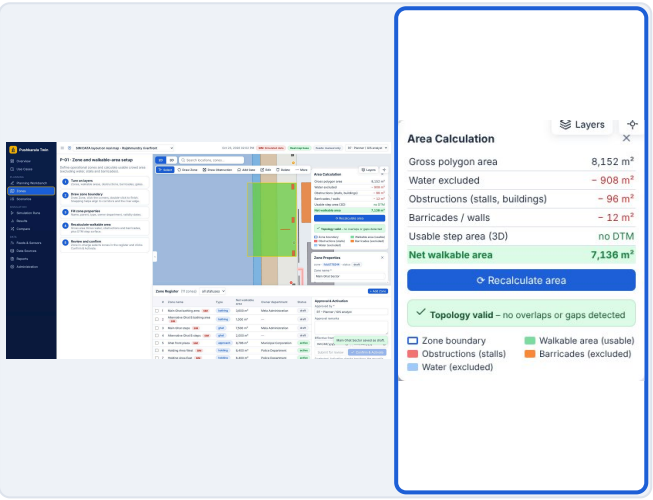
R7 clicks Draw Zone and traces the boundary on the map; snapping to corridors and the river edge is on, overlaps are prevented.

3 Fill zone properties



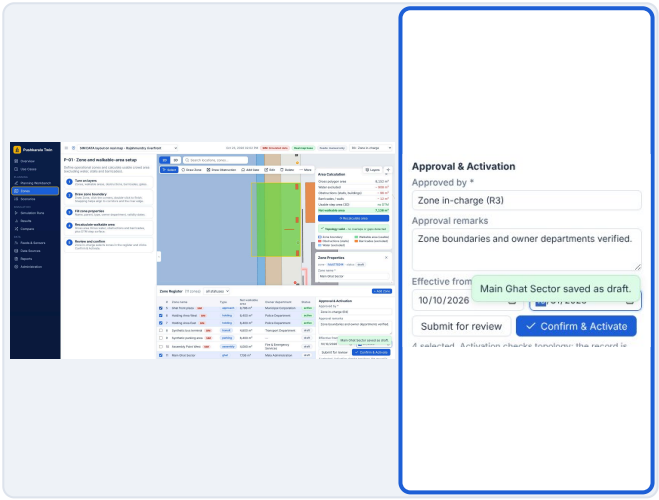
R7 fills name, parent zone, type, owner department, validity dates and description.

4 Recalculate walkable area



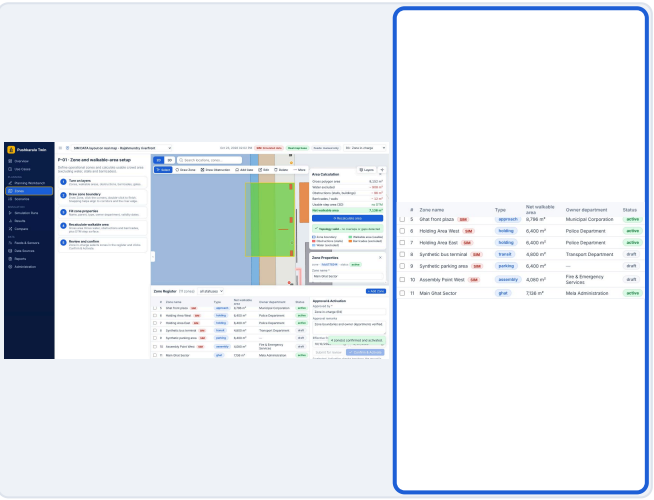
Net walkable area = gross area − water − obstructions (stalls) − barricades + DTM step surface; topology is checked.

5 Review and confirm



R3 reviews the zone register and owner departments, sets the effective period and clicks Confirm & Activate.

6 Zones active



The selected zones are active; the approval record (approver, remarks, period) is stored with each zone.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

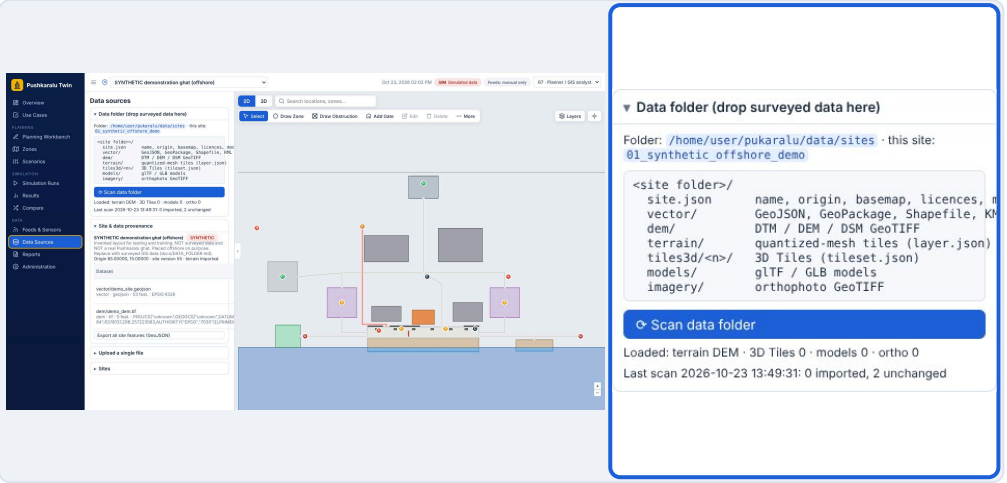
PT **Pushkaralu Twin**
User journey

P-02 Load surveyed GIS, DTM and 3D models

Copy surveyed vectors, DTM/DEM, 3D Tiles, glTF models and orthophotos into data/sites/<site>/ and load them.

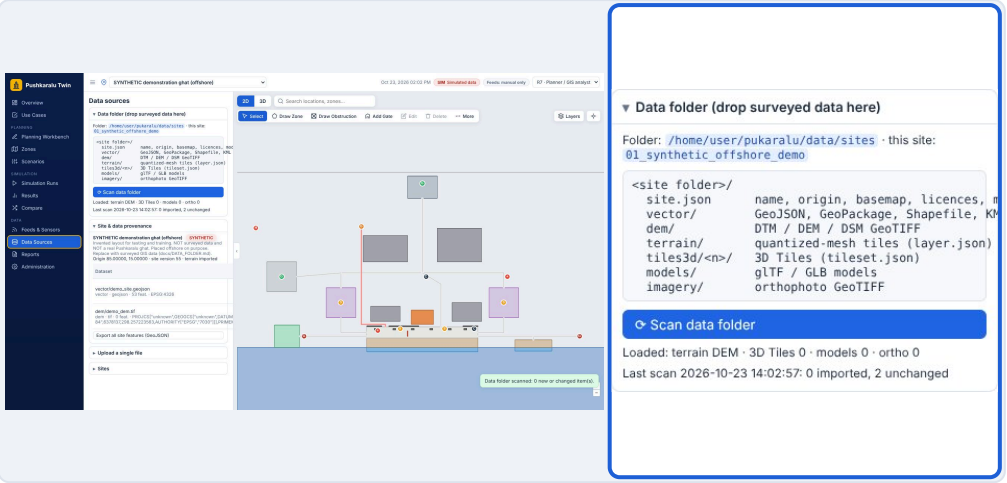
R7 · Planner / GIS analyst | R1 · Mela administrator

1 Data folder



Each site has a folder under data/sites with vector/, dem/, terrain/, tiles3d/, models/, imagery/ and an optional site.json manifest.

2 Scan data folder



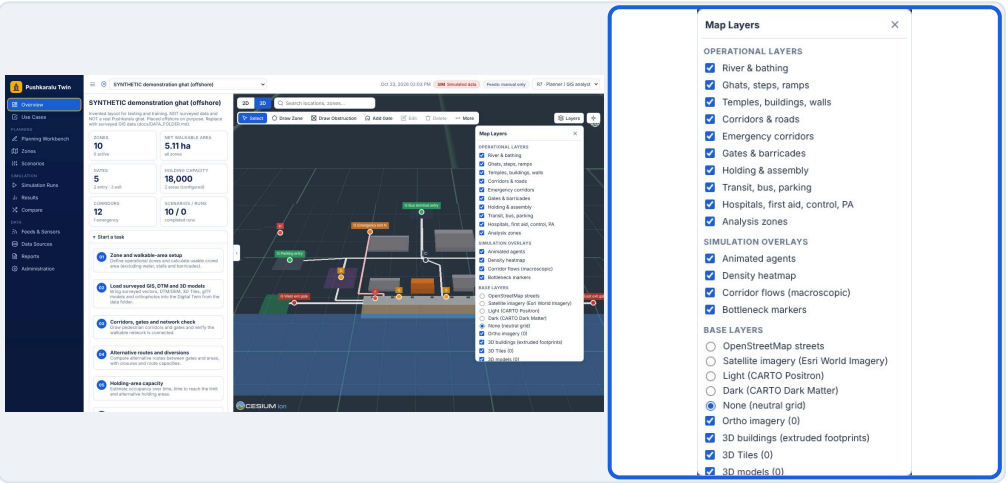
Scan data folder (or restart, or scan_data_windows.bat) imports new and changed files; unchanged files are skipped.

3 Provenance and licences



Every dataset is recorded with licence, attribution, synthetic/surveyed flags, source CRS and SHA-256.

4 Verify in 3D



The DTM drives the 3D terrain; Layers toggles ortho imagery, 3D buildings, 3D Tiles, glTF models and terrain.

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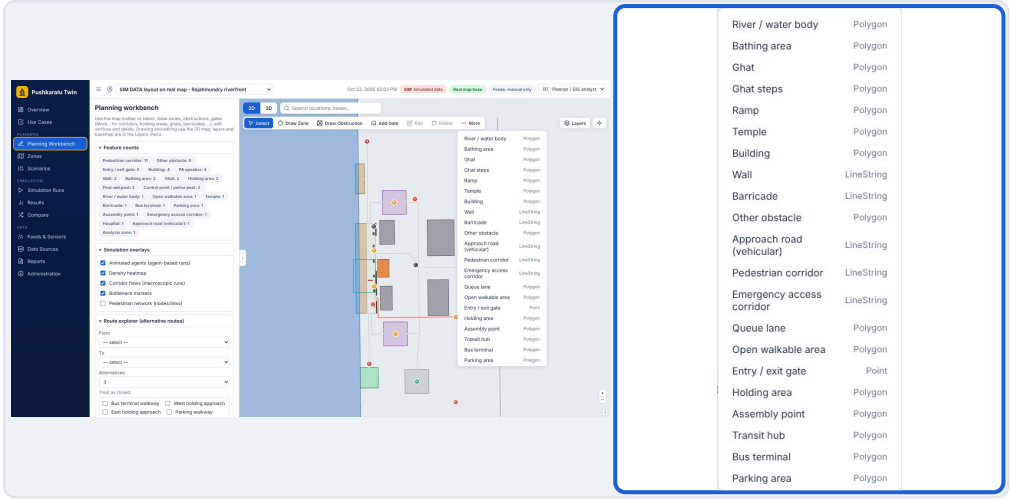
PT **Pushkaralu Twin**
User journey

P-03 Corridors, gates and network check

Draw pedestrian corridors and gates and verify that the walkable network is connected.

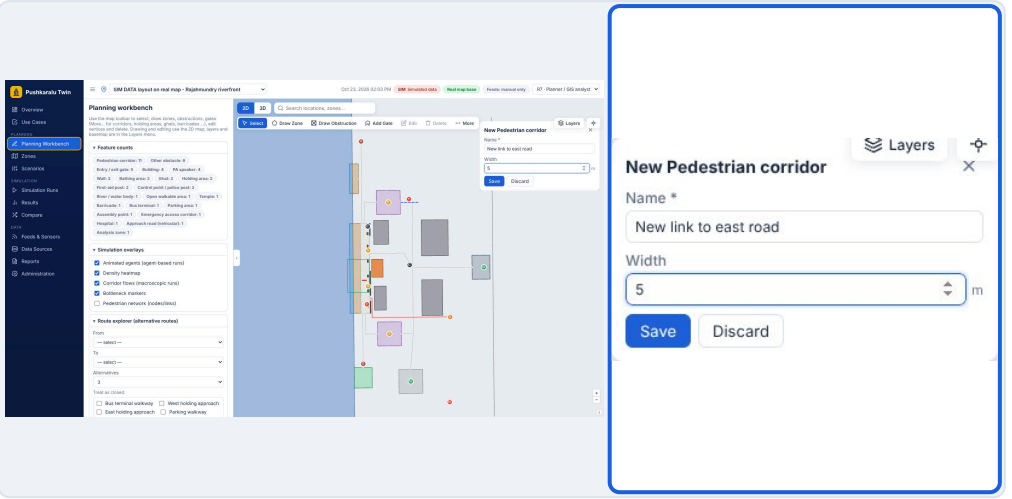
R7 · Planner / GIS analyst

1 Choose a feature type



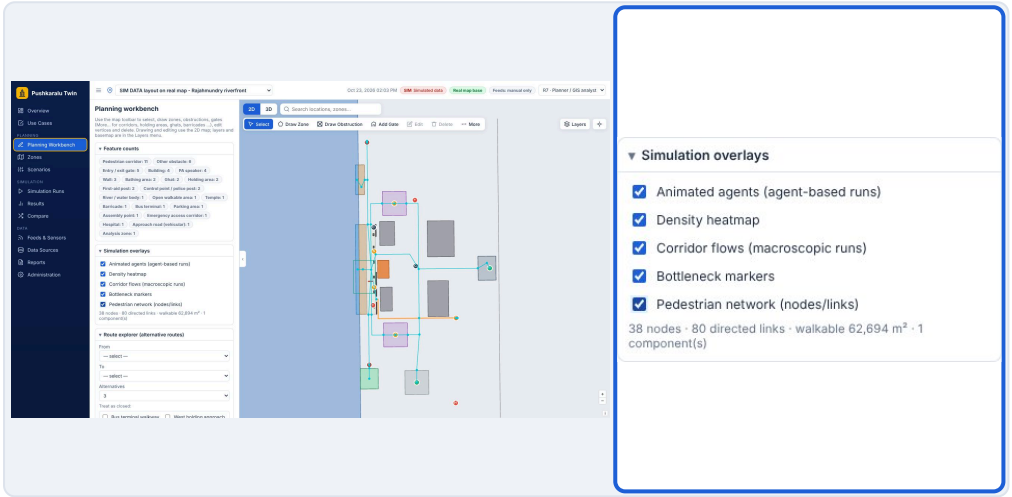
Planning Workbench: Select, Draw Zone, Draw Obstruction, Add Gate, Edit, Delete; More... lists corridors, holding areas, ghats, barricades and all other types.

2 Draw corridor and set attributes



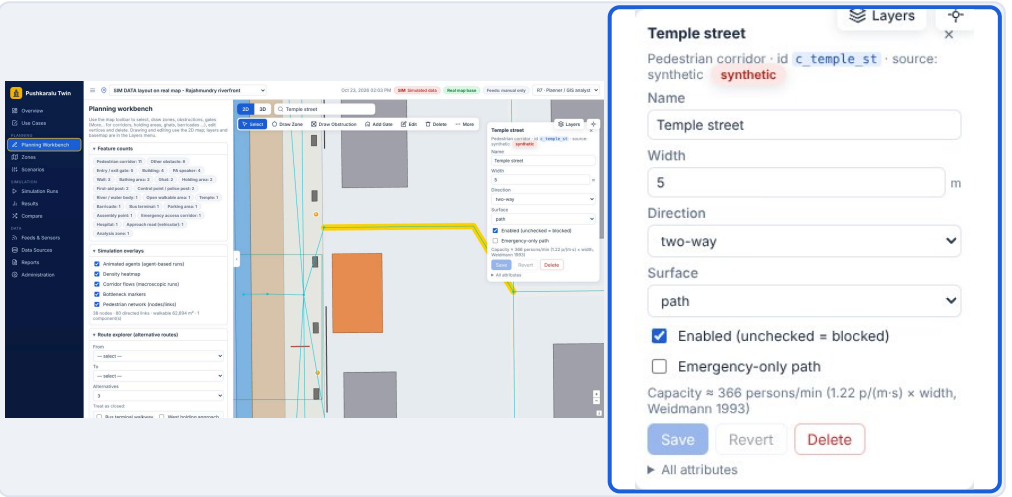
A corridor centre-line is drawn; width, direction (one-way), surface (stairs/ramps) and emergency-only are set as attributes.

3 Show the pedestrian network



The network derived from corridors, gates and areas is shown with node/link statistics and connectivity warnings.

4 Inspect and edit



Search finds a feature and flies to it; the inspector shows capacity (1.22 p/(m·s) × width) and editable attributes.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

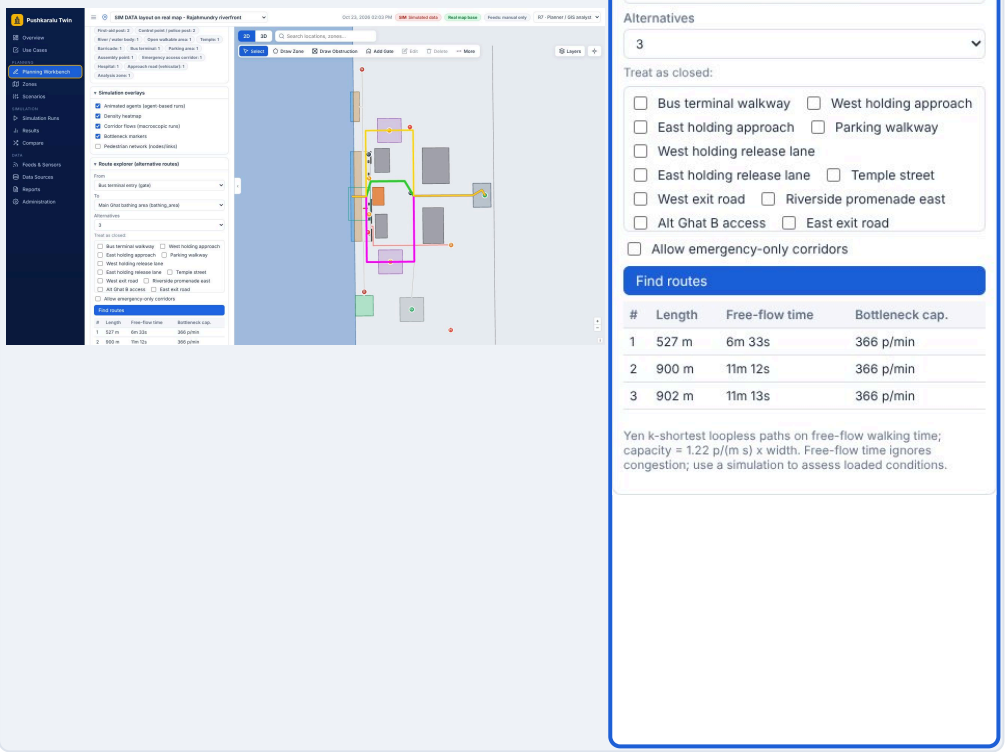
PT **Pushkaralu Twin**
User journey

P-04 Alternative routes and diversions

Compare alternative routes between gates and areas, with closures and route bottleneck capacities.

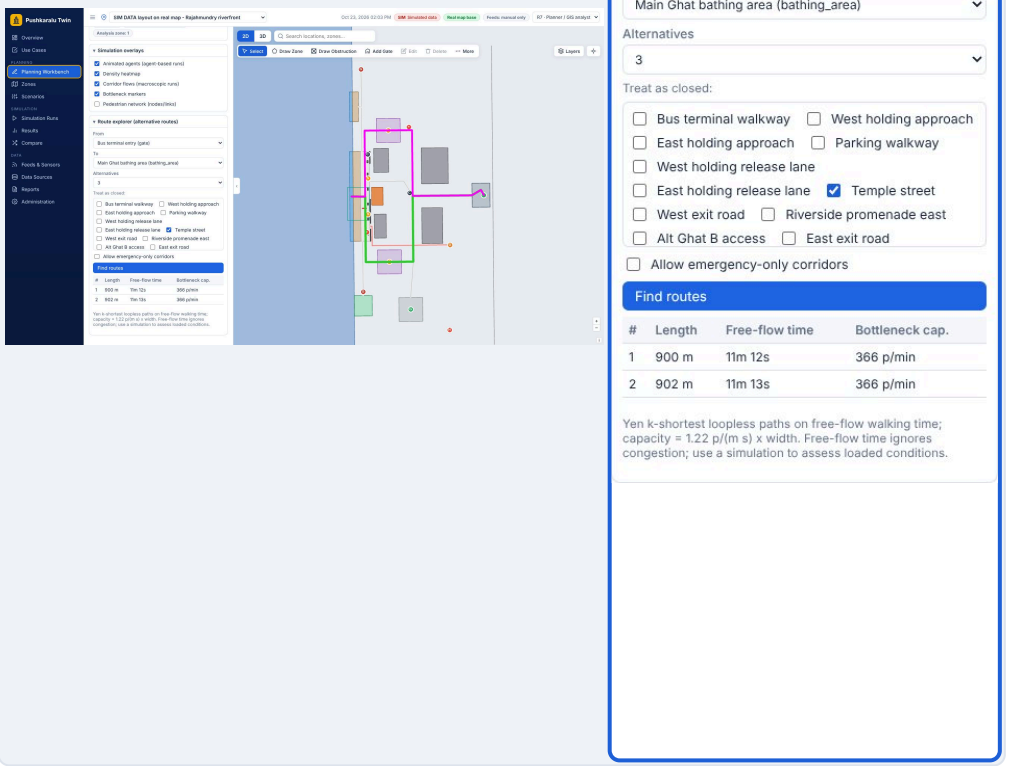
R7 · Planner / GIS analyst

1 Compare routes



Yen's k-shortest routes from the bus terminal to the bathing area: length, free-flow time and bottleneck capacity.

2 Route with a closure



With Temple street treated as closed, the alternatives use the holding-area approaches.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

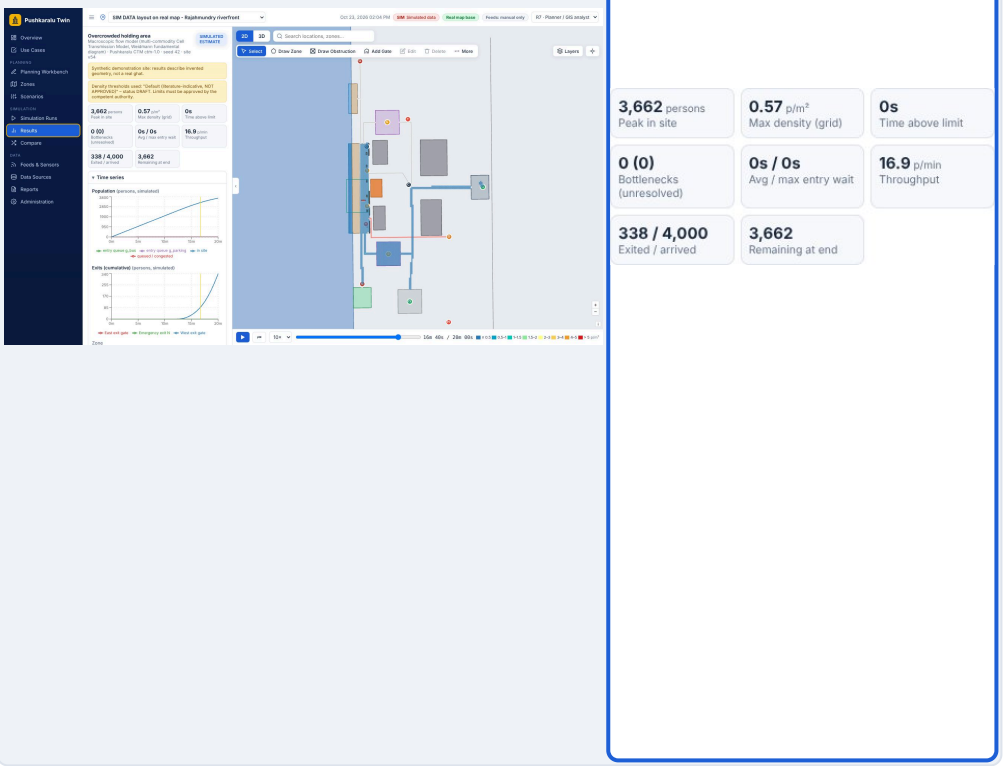
PT **Pushkaralu Twin**
User journey

P-05 Holding-area capacity

Estimate holding-area occupancy, time to reach the limit and alternatives.

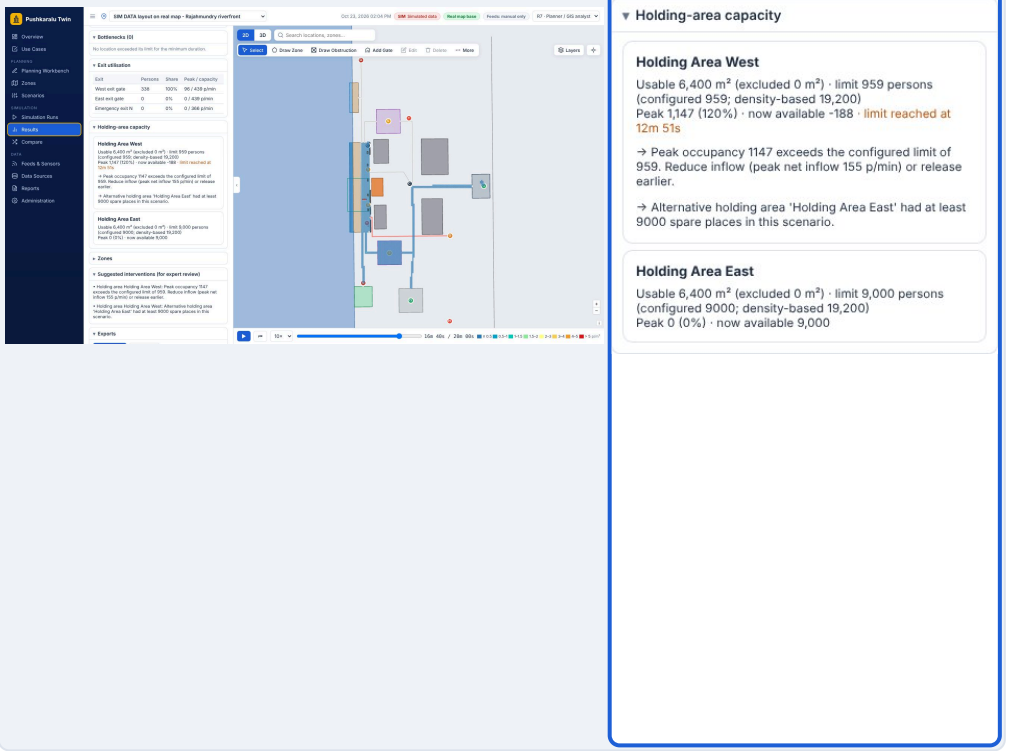
R7 · Planner / GIS analyst · R5 · Crowd safety officer

1 Overcrowded holding area



Macroscopic run of the overcrowded-holding preset: density on the map, KPIs and time series.

2 Capacity, time to limit, suggestions



Usable area excludes obstructions; limit = min(operational capacity, density limit × usable area); suggestions name alternative holding areas.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

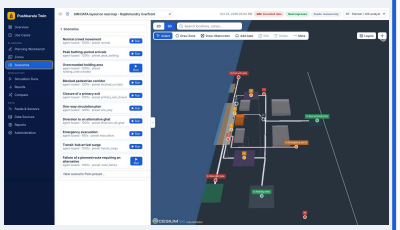
PT **Pushkaralu Twin**
User journey

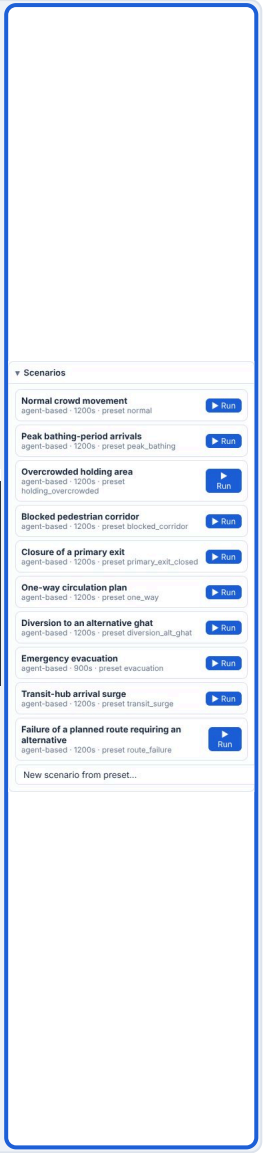
P-06 Scenario setup

Create a scenario from a preset and edit arrivals, destinations, behaviour and circulation plan.

R7 · Planner / GIS analyst

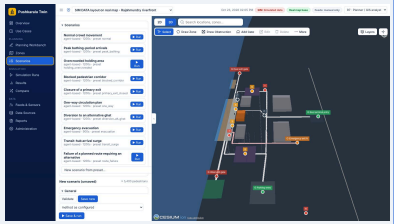
1 Scenario list and presets

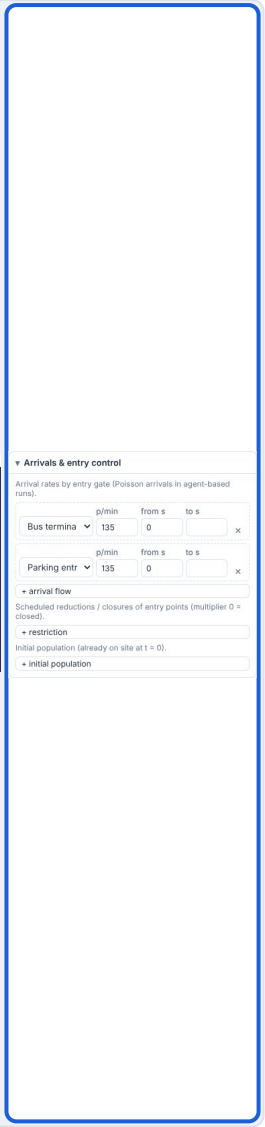




Ten presets are generated from the site's own features; every value is editable.

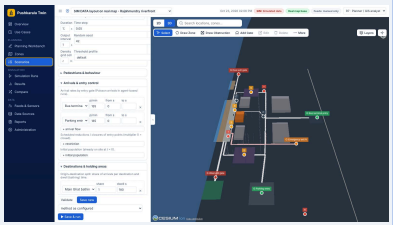
2 Edit parameters

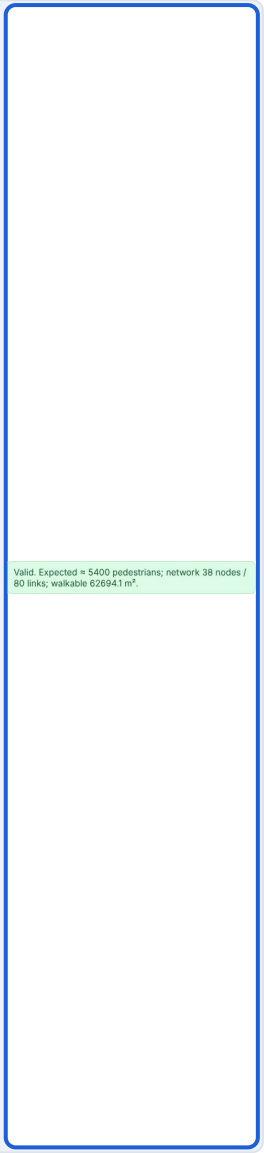




Duration, time step, method, arrivals by gate, entry restrictions, destinations, holding stops, corridors and gates, diversion, evacuation.

3 Validate





The server validates the scenario against the network (unknown gates, unreachable destinations, limits) before saving or running.

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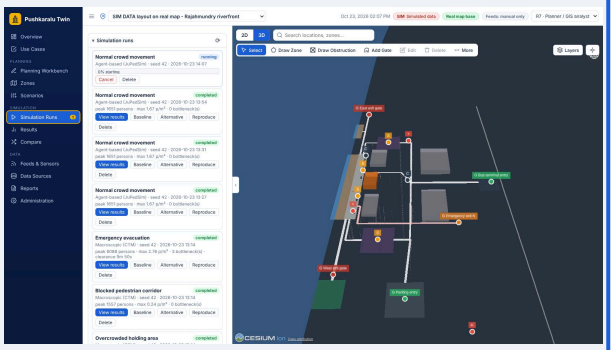
PT **Pushkaralu Twin**
User journey

P-07 Run and monitor a simulation

Run agent-based or macroscopic simulations in the background and follow their progress.

R7 · Planner / GIS analyst

1 Progress



Normal crowd movement running
Agent-based (JuPedSim) · seed 42 · 2026-10-23 14:07
0% starting
Cancel Delete

Normal crowd movement completed
Agent-based (JuPedSim) · seed 42 · 2026-10-23 13:54
peak 1651 persons · max 1.67 p/m² · 0 bottleneck(s)
View results Baseline Alternative Reproduce
Delete

Normal crowd movement completed
Agent-based (JuPedSim) · seed 42 · 2026-10-23 13:31
peak 1651 persons · max 1.67 p/m² · 0 bottleneck(s)
View results Baseline Alternative Reproduce
Delete

Normal crowd movement completed
Agent-based (JuPedSim) · seed 42 · 2026-10-23 13:27
peak 1651 persons · max 1.67 p/m² · 0 bottleneck(s)
View results Baseline Alternative Reproduce
Delete

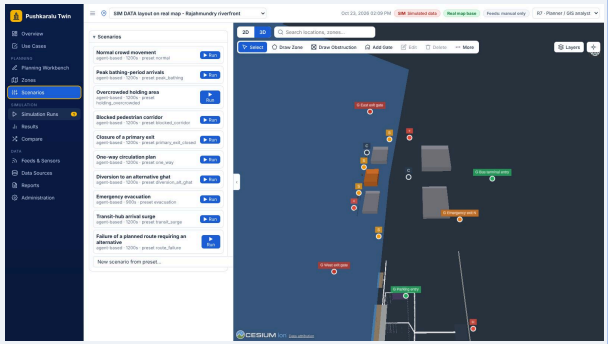
Emergency evacuation completed
Macroscopic (CTM) · seed 42 · 2026-10-23 13:14
peak 6088 persons · max 2.76 p/m² · 3 bottleneck(s) · clearance 9m 50s
View results Baseline Alternative Reproduce
Delete

Blocked pedestrian corridor completed
Macroscopic (CTM) · seed 42 · 2026-10-23 13:14
peak 1557 persons · max 0.24 p/m² · 0 bottleneck(s)
View results Baseline Alternative Reproduce
Delete

Overcrowded holding area completed
Macroscopic (CTM) · seed 42 · 2026-10-23 13:14
peak 3662 persons · max 0.57 p/m² · 0 bottleneck(s)
View results Baseline Alternative Reproduce

Runs execute in worker processes; progress is streamed to the browser (server-sent events). Runs can be cancelled.

2 Completed runs



Normal crowd movement
agent-based · 1200s · preset normal ▶ Run

Peak bathing-period arrivals
agent-based · 1200s · preset peak_bathing ▶ Run

Overcrowded holding area
agent-based · 1200s · preset holding_overcrowded ▶ Run

Blocked pedestrian corridor
agent-based · 1200s · preset blocked_corridor ▶ Run

Closure of a primary exit
agent-based · 1200s · preset primary_exit_closed ▶ Run

One-way circulation plan
agent-based · 1200s · preset one_way ▶ Run

Diversion to an alternative ghat
agent-based · 1200s · preset diversion_alt_ghat ▶ Run

Emergency evacuation
agent-based · 900s · preset evacuation ▶ Run

Transit-hub arrival surge
agent-based · 1200s · preset transit_surge ▶ Run

Failure of a planned route requiring an alternative
agent-based · 1200s · preset route_failure ▶ Run

Each run records method, engine version, seed, data snapshot and thresholds; Reproduce re-runs it exactly.

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PT

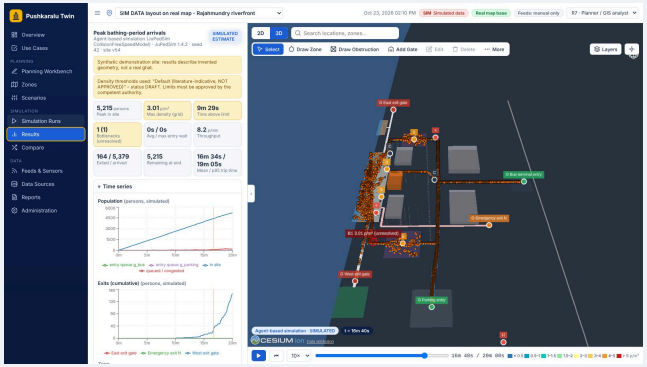
Pushkaralu Twin
User journey

P-08 Density, queues and bottlenecks

Replay agents and the density heatmap; inspect zone densities, queues and bottlenecks.

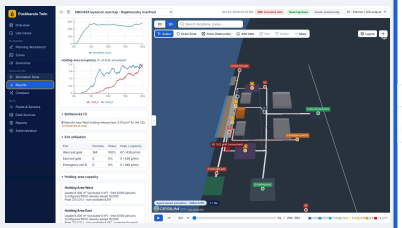
R5 · Crowd safety officer

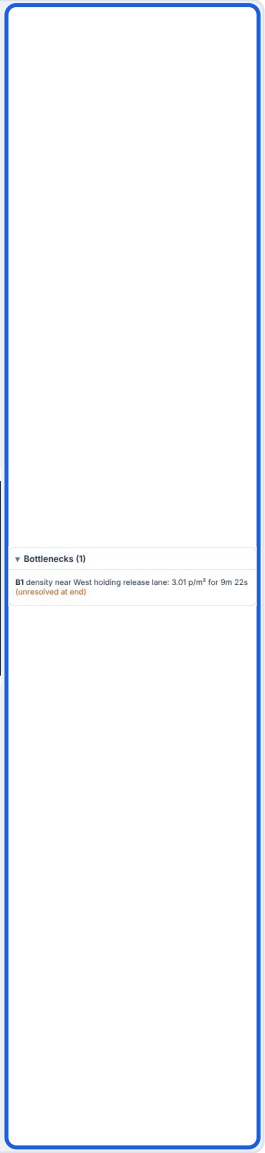
1 Replay in 3D



Agent-based peak-arrival run at t = 16 min 40 s: agents, density heatmap and bottleneck markers over the real map.

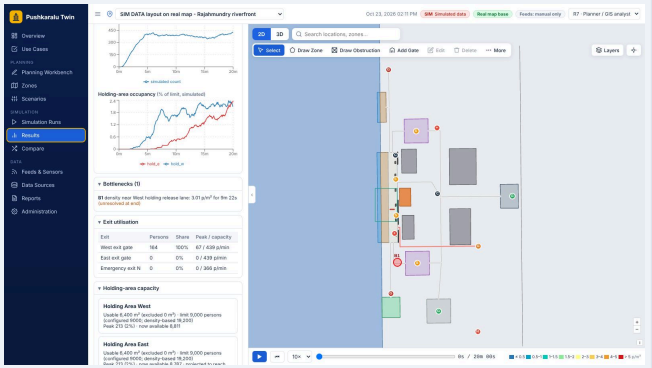
2 Bottlenecks





Bottlenecks = locations above the configured limit for at least the minimum duration; unresolved ones are flagged.

3 2D density map



The same frame in the 2D planning map.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

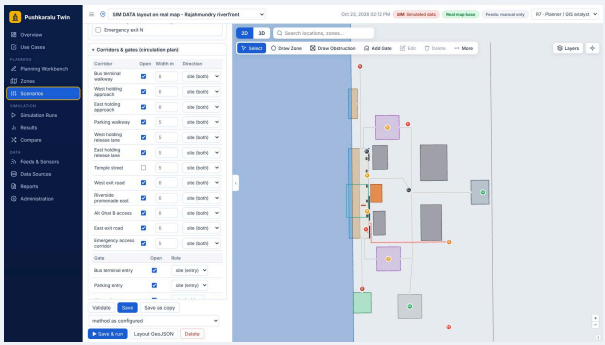
PT **Pushkaralu Twin**
User journey

P-09 Blocked corridor and route failure

Block a corridor (or let a route fail mid-run) and re-run with re-routing.

R7 · Planner / GIS analyst | R5 · Crowd safety officer

1 Disable a corridor



▾ Corridors & gates (circulation plan)

Corridor	Open	Width m	Direction
Bus terminal walkway	☑	8	site (both)
West holding approach	☑	6	site (both)
East holding approach	☑	6	site (both)
Parking walkway	☑	5	site (both)
West holding release lane	☑	5	site (both)
East holding release lane	☑	5	site (both)
Temple street	☐	5	site (both)
West exit road	☑	6	site (both)
Riverside promenade east	☑	6	site (both)
Alt Ghat B access	☑	6	site (both)
East exit road	☑	6	site (both)
Emergency access corridor	☑	5	site (both)

Gate	Open	Role
Bus terminal entry	☑	site (entry)
Parking entry	☑	site (entry)

Validate | Save | Save as copy

method as configured

▶ Save & run | Layout GeoJSON | Delete

Corridors can be closed, narrowed or made one-way per scenario; gates can be closed or change role.

2 Re-routed flows



1,557 persons Peak in site	**0.24** p/m² Max density (grid)	**0s** Time above limit
0 (0) Bottlenecks (unresolved)	**0s / 0s** Avg / max entry wait	**12.2** p/min Throughput
243 / 1,800 Exited / arrived	**1,557** Remaining at end	

Macroscopic result with the corridor closed: computed corridor flows show the re-routed pedestrians.

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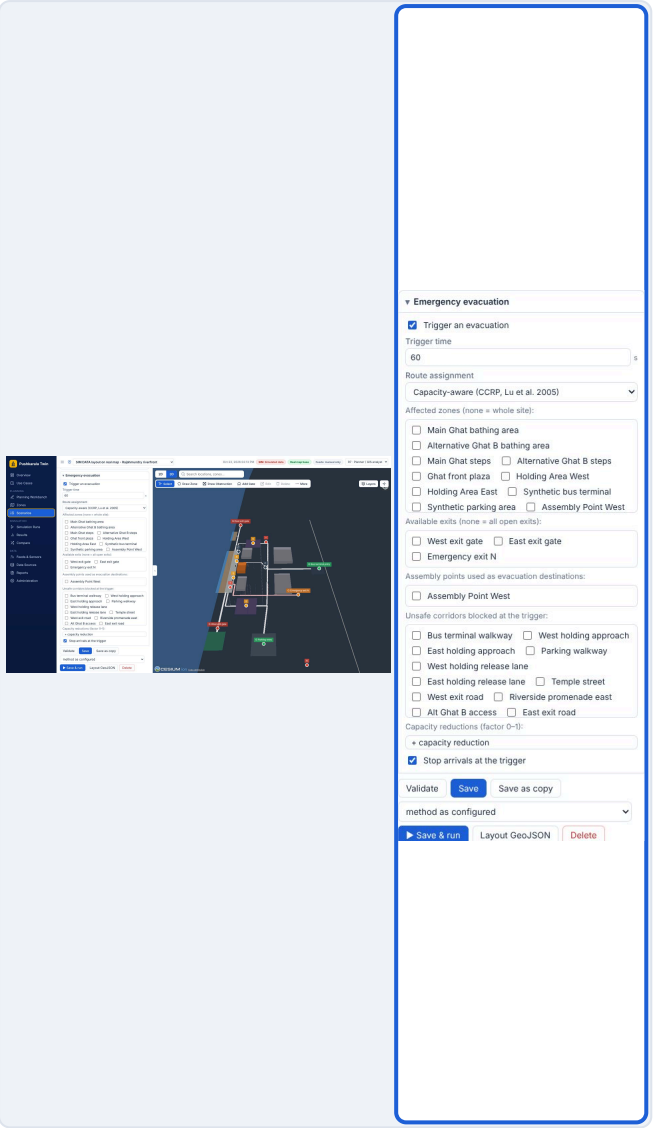
PT **Pushkaralu Twin**
User journey

P-10 Emergency evacuation

Simulate an evacuation with chosen exits, closures and routing strategy; read clearance times.

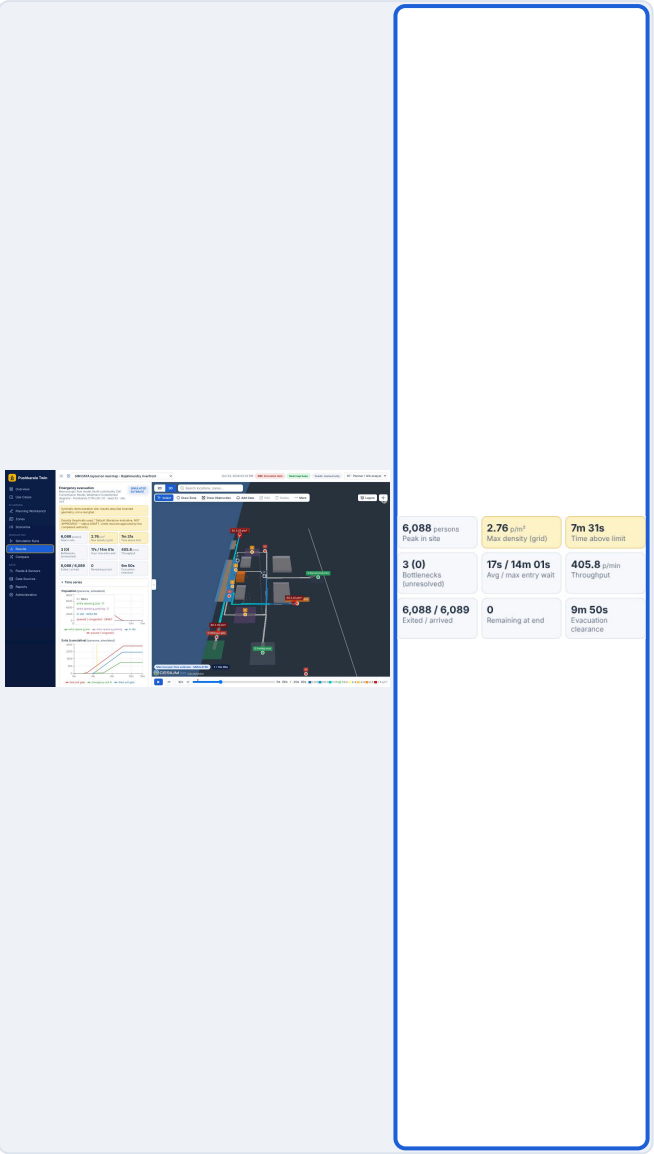
R5 · Crowd safety officer | R1 · Mela administrator

1 Configure evacuation



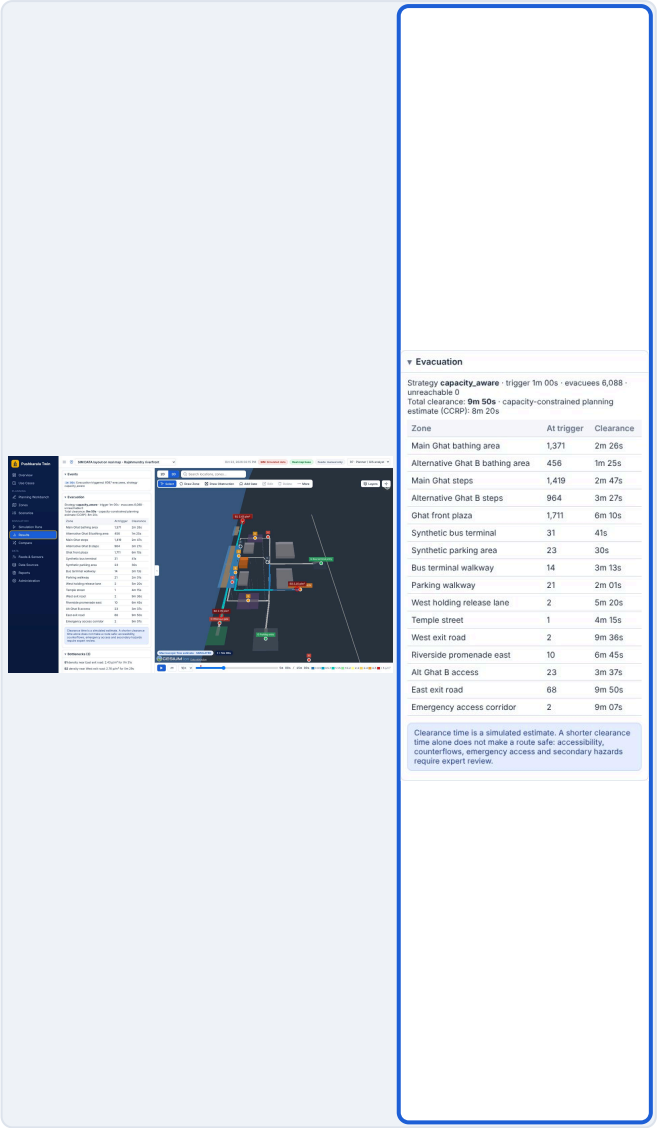
Trigger time, affected zones, exits and assembly points, routing (nearest / capacity-aware CCRP / assigned), closures and capacity reductions.

2 Evacuation flows



Macroscopic evacuation at t = 5 min: corridor flows, bottlenecks and clearance KPIs.

3 Clearance by zone



Total and per-zone clearance, unreachable evacuees and the CCRP planning estimate. A short clearance time alone does not make a route safe.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

PT

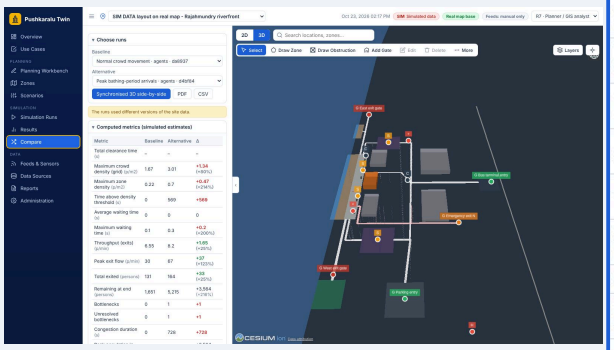
Pushkaralu Twin
User journey

P-11 Compare baseline and alternative

Compare two runs with computed metrics and synchronised 3D views.

R5 · Crowd safety officer

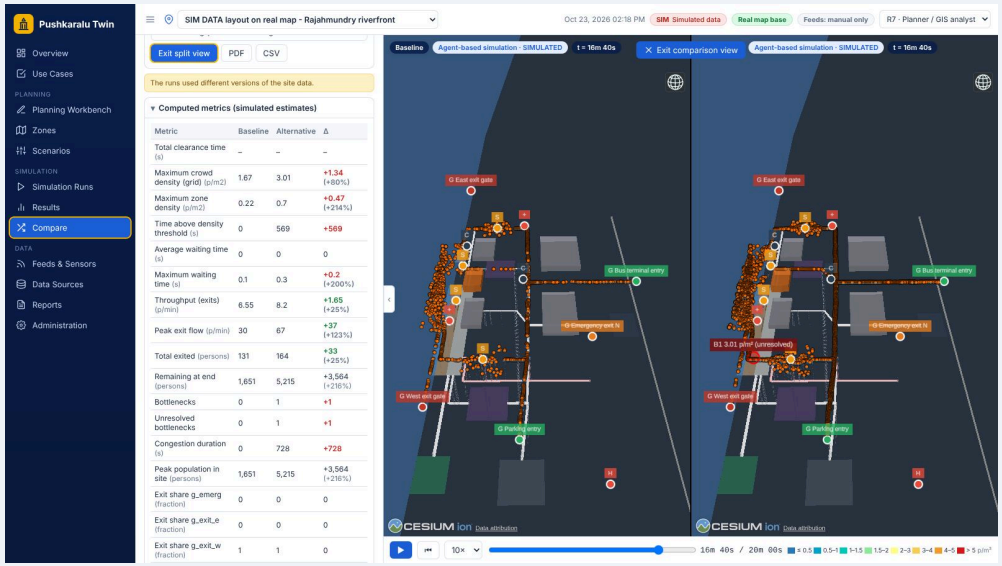
1 Metric differences



Metric	Baseline	Alternative	Δ
Total clearance time (s)	–	–	–
Maximum crowd density (grid) (p/m2)	1.67	3.01	+1.34 (+80%)
Maximum zone density (p/m2)	0.22	0.7	+0.47 (+214%)
Time above density threshold (s)	0	569	+569
Average waiting time (s)	0	0	0
Maximum waiting time (s)	0.1	0.3	+0.2 (+200%)
Throughput (exits) (p/min)	6.55	8.2	+1.65 (+25%)
Peak exit flow (p/min)	30	67	+37 (+123%)
Total exited (persons)	131	164	+33 (+25%)
Remaining at end (persons)	1,651	5,215	+3,564 (+216%)
Bottlenecks	0	1	+1
Unresolved bottlenecks	0	1	+1
Congestion duration (s)	0	728	+728
Peak population in site (persons)	1,651	5,215	+3,564 (+216%)
Exit share g_emerg (fraction)	0	0	0
Exit share g_exit_e (fraction)	0	0	0
Exit share g_exit_w (fraction)	1	1	0

Differences are computed from the two runs (clearance, densities, time above limit, waiting, throughput, exit use, bottlenecks).

2 Synchronised 3D



Two 3D viewers on a shared timeline with synchronised cameras.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

PT

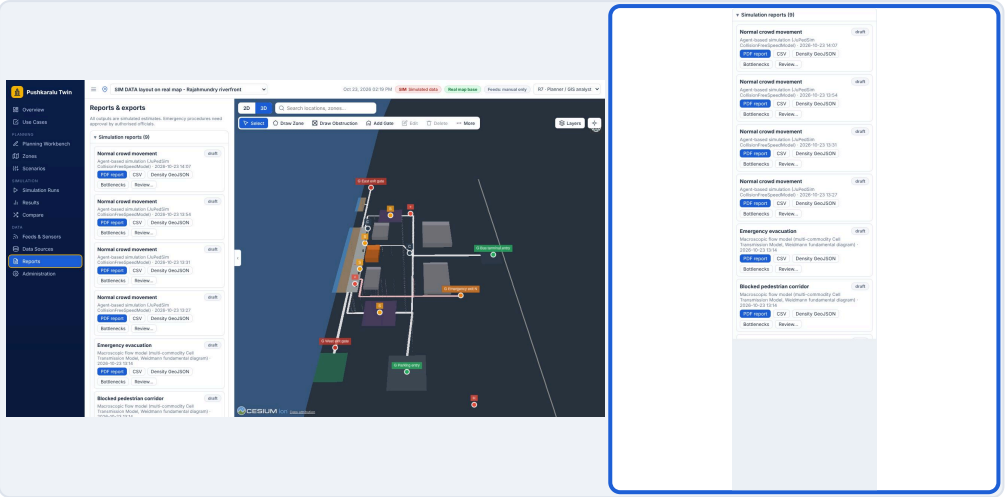
Pushkaralu Twin
User journey

P-12 Reports, SOP and drill package

Download PDF/CSV/GeoJSON outputs and record review and approval.

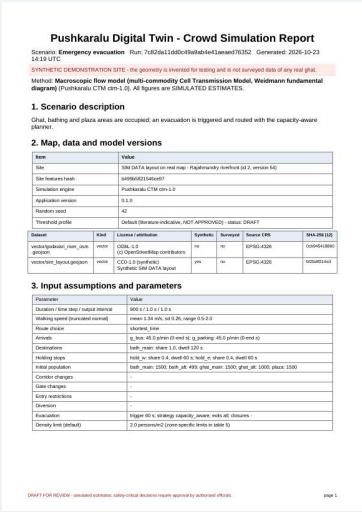
R5 · Crowd safety officer | R1 · Mela administrator

1 Reports and exports



PDF report, CSV, density and bottleneck GeoJSON per run; comparison report; circulation layouts as GeoJSON.

2 PDF report – inputs and provenance



Pushkaralu Digital Twin - Crowd Simulation Report

Scenario: **Emergency evacuation** Run: 7c82da11dd0c49a8ab4e41aeaed76352 Generated: 2026-10-23 14:19 UTC

SYNTHETIC DEMONSTRATION SITE - the geometry is invented for testing and is not surveyed data of any real ghat.

Method: **Macroscopic flow model (multi-commodity Cell Transmission Model, Weidmann fundamental diagram)** (Pushkaralu CTM ctm-1.0). All figures are SIMULATED ESTIMATES.

1. Scenario description

Ghat, bathing and plaza areas are occupied; an evacuation is triggered and routed with the capacity-aware planner.

2. Map, data and model versions

Item	Value
Site	SIM DATA layout on real map - Rajahmundry riverfront (id 2, version 54)
Site features hash	b499b82154fce97
Simulation engine	Pushkaralu CTM ctm-1.0
Application version	0.1.0
Random seed	42
Threshold profile	Default (literature-indicative, NOT APPROVED) - status: DRAFT

Dataset	Kind	License / attribution	Synthetic	Surveyed	Source CRS	SHA-256 (12)
vector/geodetic/_river_ctcm_geojson	vector	CCBAL-1.0 (c) OpenStreetMap contributors	no	no	EPSG:4326	0b99d41d89b0
vector/sim_layout.geojson	vector	CCO-1.0 (synthetic) Synthetic SIM DATA layout	yes	no	EPSG:4326	9f0bd81d4a3

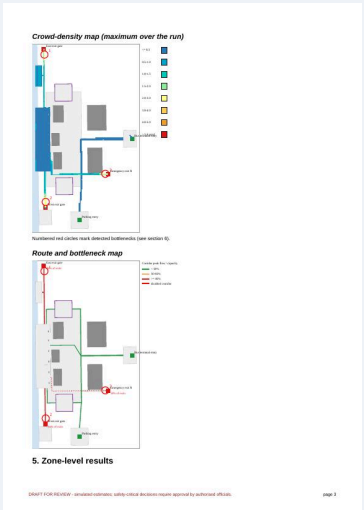
3. Input assumptions and parameters

Parameter	Value
Duration / time step / output interval	900 s / 1.0 s / 1.0 s
Walking speed (truncated normal)	mean 1.34 m/s, sd 0.26, range 0.5-2.0
Route choice	shortest_time
Arrivals	g_bus: 45.0 p/min (0-end s); g_parking: 45.0 p/min (0-end s)
Destinations	bath_main: share 1.0, dwell 120 s
Holding stops	hold_w: share 0.4, dwell 60 s; hold_e: share 0.4, dwell 60 s
Initial population	bath_main: 1000; bath_ult: 400; ghat_main: 1500; ghat_ult: 1000; plaza: 1500
Corridor changes	-
Gate changes	-
Entry restrictions	-
Diversion	-
Evacuation	trigger 60 s; strategy capacity_aware; exits all; closures -
Density limit (default)	2.0 persons/m2 (zone-specific limits in table 3)

DRAFT FOR REVIEW - simulated estimates; safety-critical decisions require approval by authorised officials. page 1

Scenario description, map/data versions, engine and seed, data licences and input assumptions.

3 PDF report – maps



Crowd-density map (maximum over the run)

Numbered red circles mark detected bottlenecks (see section 6).

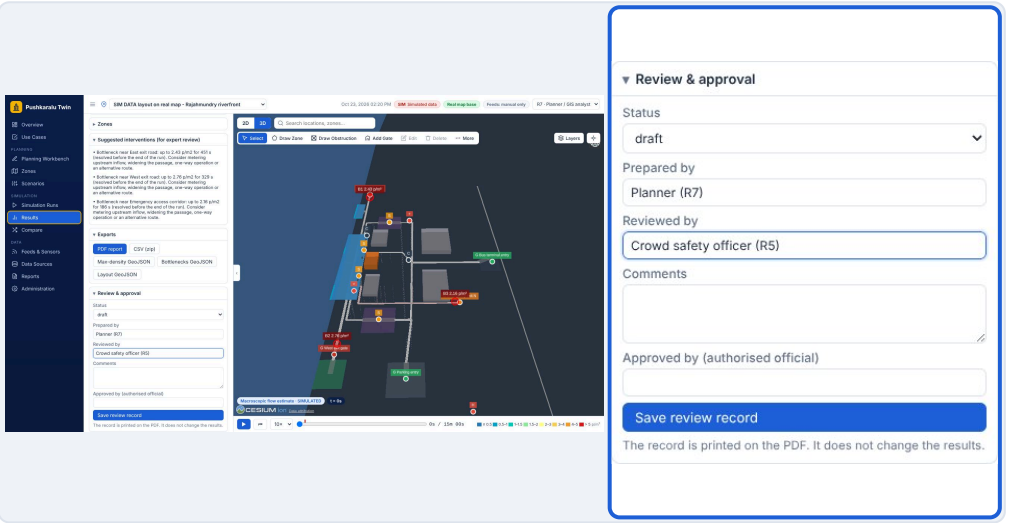
Route and bottleneck map

5. Zone-level results

DRAFT FOR REVIEW - simulated estimates; safety-critical decisions require approval by authorised officials. page 3

Density map and route/bottleneck map computed from the run.

4 Review and approval record



Review status, reviewer and approver are stored with the run and printed in the report's approval block.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

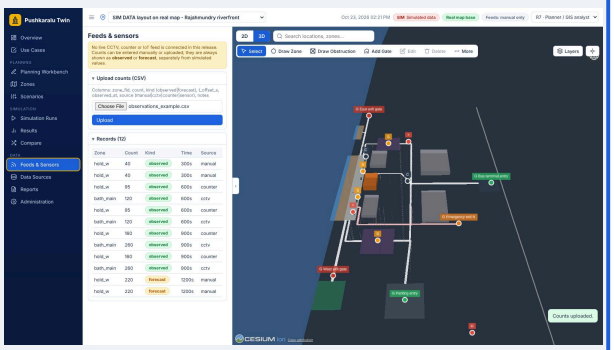
PT **Pushkaralu Twin**
User journey

P-13 Observed and forecast counts

Upload manual or counter counts and compare them with simulated values.

R7 · Planner / GIS analyst

1 Upload counts

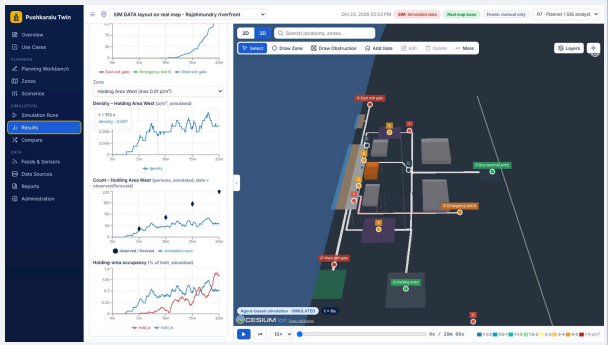


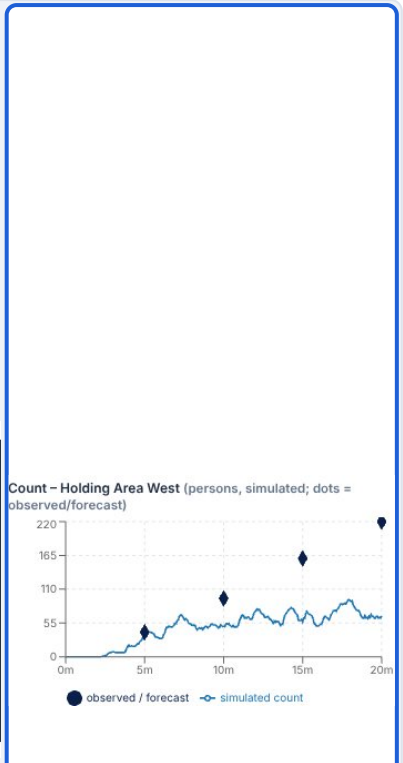
▾ Records (12)

Zone	Count	Kind	Time	Source
hold_w	40	observed	300s	manual
hold_w	40	observed	300s	manual
hold_w	95	observed	600s	counter
bath_main	120	observed	600s	cctv
hold_w	95	observed	600s	counter
bath_main	120	observed	600s	cctv
hold_w	160	observed	900s	counter
bath_main	260	observed	900s	cctv
hold_w	160	observed	900s	counter
bath_main	260	observed	900s	cctv
hold_w	220	forecast	1200s	manual
hold_w	220	forecast	1200s	manual

Counts are stored as observed or forecast with their source; no live feeds are connected in this release.

2 Compare with simulation





Observed/forecast values appear as separate markers on the simulated zone count (example values).

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.

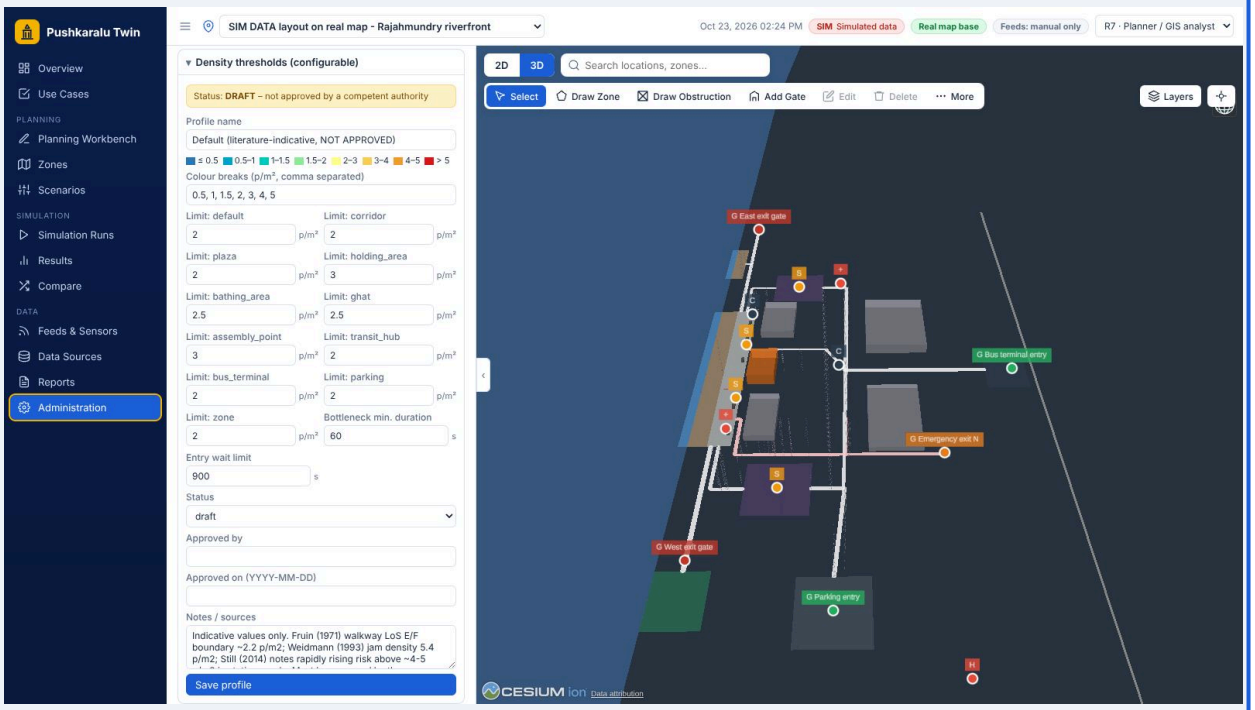
PT **Pushkaralu Twin**
User journey

P-14 Density thresholds approval

Configure density limits and record approval by the competent authority.

R1 · Mela administrator | R5 · Crowd safety officer

1 Thresholds and approval status





Colour breaks and limits per zone type; the default profile is literature-indicative and marked NOT APPROVED.

SIM DATA – screenshots show synthetic demonstration data and simulated estimates, not surveyed Pushkaralu data. Safety-critical decisions require approval by authorised officials.